

MADRID TORREJON AB, SpainWMO: **082270**Lat: **40.4833N**Lon: **3.4436W**Elev: **618**StdP: **94.12**Time Zone: **1.00 (EUC)**Period: **97-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.8	-3.1	-10.0	1.7	3.2	-7.9	2.1	4.0	12.0	7.3	10.8	7.7	1.9	30	0.486

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.4	36.9	19.3	35.2	18.6	34.0	18.1	21.6	34.1	20.3	31.6	19.3	30.4	3.7	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.1	13.1	26.5	16.1	12.3	25.0	15.1	11.6	23.8	65.9	33.9	61.5	31.6	57.5	30.6	25.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.0	8.6	7.5	-7.3	38.8	2.1	1.2	-8.8	39.6	-10.1	40.3	-11.2	41.0	-12.8	41.9		
			-8.1	21.6	2.0	1.8	-9.5	22.9	-10.7	24.0	-11.8	25.0	-13.3	26.3		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	14.5	5.6	6.6	9.8	12.5	16.5	21.9	25.0	24.8	20.5	15.3	9.1	6.0	(6)
(7)		DBStd	7.63	2.91	2.80	3.05	3.28	3.63	3.66	2.87	2.67	3.08	3.23	3.28	2.88	(7)
(8)		HDD10.0	476	139	98	41	13	2	0	0	0	3	54	126		(8)
(9)		HDD18.3	2031	395	328	264	177	79	11	1	14	103	277	381		(9)
(10)		CDD10.0	2122	2	3	35	87	202	357	466	459	314	166	27	3	(10)
(11)		CDD18.3	636	0	0	0	1	21	118	208	201	79	8	0	0	(11)
(12)		CDH23.3	8841	0	0	6	68	398	1728	2914	2631	959	137	0	0	(12)
(13)		CDH26.7	4383	0	0	0	7	119	853	1609	1404	363	27	0	0	(13)
(14)	Wind	WSAvg	2.9	2.7	3.2	3.2	3.2	3.0	3.0	2.9	2.6	2.5	2.7	2.4	(14)	
(15)	Precipitation	PrecAvg	401	33	31	30	43	50	22	8	9	27	60	49	40	(15)
(16)		PrecMax	571	95	92	98	100	112	66	26	34	65	123	175	135	(16)
(17)		PrecMin	233	0	1	0	11	1	1	0	0	2	7	5	2	(17)
(18)		PrecStd	83	26	22	24	22	34	18	8	9	15	34	36	37	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.2	19.9	24.1	27.8	32.2	38.0	38.2	35.2	29.2	21.1	16.1	(19)	
(20)			MCWB	9.9	10.3	12.8	15.0	17.1	20.0	19.2	20.2	18.0	15.7	13.6	10.4	(20)
(21)		2%	DB	14.2	17.0	21.8	25.2	29.9	35.2	36.9	36.2	32.1	27.0	18.8	14.2	(21)
(22)			MCWB	9.3	9.3	11.3	13.9	15.9	19.2	19.1	19.2	17.7	15.1	12.4	9.7	(22)
(23)		5%	DB	12.8	14.9	19.8	23.0	27.2	33.8	35.2	35.0	30.2	24.8	16.7	12.9	(23)
(24)			MCWB	8.8	8.5	10.6	13.1	15.0	18.2	18.5	18.6	16.9	14.5	11.5	9.4	(24)
(25)		10%	DB	11.0	12.9	17.1	20.1	25.2	31.8	34.0	33.8	28.9	22.1	14.9	11.1	(25)
(26)			MCWB	7.8	8.2	9.9	11.7	14.3	17.4	18.0	18.1	16.6	14.0	10.8	8.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.0	12.2	13.8	16.1	18.6	22.2	22.6	23.0	20.2	18.1	15.3	12.8	(27)
(28)			MCDWB	14.4	15.8	22.4	25.7	29.5	35.4	35.7	35.6	30.4	25.5	18.6	14.5	(28)
(29)		2%	WB	10.6	10.7	12.5	14.7	16.9	20.5	21.4	21.8	19.1	17.0	13.7	11.0	(29)
(30)			MCDWB	13.0	15.1	19.2	23.2	27.3	32.7	34.0	34.0	28.9	22.6	17.0	13.1	(30)
(31)		5%	WB	9.4	9.5	11.5	13.6	16.0	19.2	20.0	20.4	18.3	16.1	12.7	9.9	(31)
(32)			MCDWB	11.7	13.6	17.3	21.4	24.9	31.0	31.9	30.9	27.7	21.1	15.5	12.0	(32)
(33)		10%	WB	8.4	8.6	10.5	12.5	15.1	18.0	18.8	19.2	17.4	15.2	11.5	8.9	(33)
(34)			MCDWB	10.5	12.3	15.7	18.7	23.0	29.2	30.9	29.8	26.1	19.9	14.2	11.1	(34)
(35)	Mean Daily Temperature Range	MDBR	10.1	11.7	12.6	12.7	13.9	15.7	16.4	15.9	14.5	12.1	10.2	10.2	(35)	
(36)		5% DB	MCDBR	13.5	16.5	17.9	17.9	18.1	18.6	17.7	17.9	17.2	16.6	12.9	12.6	(36)
(37)			MCWBR	8.6	9.7	9.1	8.1	7.3	7.3	6.7	7.3	7.4	7.4	7.3	8.3	(37)
(38)		5% WB	MCDBR	10.1	13.1	14.4	15.5	15.9	18.0	17.3	16.5	15.4	11.6	9.6	9.6	(38)
(39)			MCWBR	7.0	8.2	7.9	7.3	6.8	8.1	8.1	8.3	7.3	5.8	5.9	6.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.289	0.303	0.338	0.347	0.361	0.364	0.347	0.356	0.345	0.332	0.304	0.289	(40)	
(41)		taud	2.512	2.483	2.404	2.386	2.373	2.384	2.407	2.384	2.427	2.481	2.516	2.521	(41)	
(42)		Ebn at Noon	883	918	915	926	915	909	922	905	895	871	855	853	(42)	
(43)		Edh at Noon	74	88	106	116	120	119	115	115	103	87	73	68	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.11	3.02	4.25	5.41	6.48	7.41	7.73	6.70	5.15	3.45	2.32	1.89	(44)	
(45)		RadStd	0.30	0.37	0.50	0.50	0.53	0.33	0.23	0.22	0.21	0.32	0.31	0.22	(45)	

Historical Trends

		DBAvg	Heating		Cooling
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